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Conformable dressing

Abstract

When thin-film dressings are used to secure a device, the dressing might need to stretch in some areas to accommodate varying sizes of a device, while remaining stiffer in some areas to limit stretching of the materials. Excessive stretching of the materials can cause the dressing to peel away from the skin or cause the underlying device to not be sufficiently secured. The disclosed dressing can be used to secure a device and includes perforations near the device securement area to allow for isolated regions of adjustability to accommodate a secure fit between the dressing and device.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a national stage filing under 35 U.S.C. 371 of PCT/IB2020/052552, filed Mar. 19, 2020, which claims the benefit of Provisional Application No. 62/827,400, filed Apr. 1, 2019, the disclosure of which is incorporated by reference in its/their entirety herein.

TECHNICAL FIELD

(1) The present disclosure relates to a conformable dressing comprising a backing layer with a perforation set for enabling controlled elasticity.

BACKGROUND

(2) Transparent film dressings are widely used on skin because they protect the skin by acting as a barrier to contaminating liquids and bacteria. For example, transparent film dressings are available under a number of trade names such as TEGADERM™ (3M Company, St. Paul, MN).

(3) The polymeric films used in those dressings and drapes are conformable, i.e., the films are extremely thin, flexible and supple. They are typically supplied with a releasable protective liner covering the adhesive coated surface of the film. To prevent the adhesive coated film from

wrinkling during application, various delivery systems have been proposed to address this problem, such as those disclosed in U.S. Pat. No. 6,685,682.

(4) Thin polymeric films that are flexible and resilient are beneficial when used on skin that flexes, stretches, and retracts. However, for some applications, the high flexibility and resiliency of the thin polymeric film can move and stretch too much causing the film to peel or release from the skin. Therefore, medical dressings have been developed that include stiffer, less conformable material secured to the thin polymeric film. For example, U.S. Pat. No. 5,088,483 discloses an adhesive composite that includes a conformable backing and a permanent adhesive reinforcement around the periphery of the adhesive composite. One example of a commercially available medical dressing with a reinforcement layer is TEGADERM™ IV Advanced Dressing (3M Company, St. Paul MN).

SUMMARY

(5) When thin-film dressings are used to secure a device, the dressing might need to stretch in some areas to accommodate varying sizes of a device, while remaining stiffer in some areas to limit stretching of the materials. Excessive stretching of the materials can cause the dressing to peel away from the skin or cause the underlying device to not be sufficiently secured. The disclosed dressing can be used to secure a device and includes perforations near the device securement area to allow for isolated regions of adjustability to accommodate a secure fit between the dressing and device.

(6) In one embodiment, the dressing comprises a first major surface; second major surface, opposite the first major surface, defined by a perimeter around a body layer, wherein the second major surface comprises an adhesive; and a slot for receiving a device. The slot is an extension of the perimeter within the body layer to a terminal end. The slot is separated by a first and second portion of the body layer. In one embodiment, the dressing further comprises a first perforation set that extends at least partially through the thickness of the flexible backing layer and extending from the perimeter of the dressing at the slot to within the body layer.

(7) In one embodiment, the body layer further comprises a support material secured to the body layer. In one embodiment, the support material covers a portion of the body layer. In one embodiment, the support material extends to the perimeter. In one embodiment, the support material is less elastic than the body layer.

(8) In one embodiment, the first perforation set extends through the thickness of the support material. In one embodiment, the slot extends along a dressing slot axis. In one embodiment, the first and second body portions are symmetrical about the dressing slot axis. In one embodiment, the first perforation set is at the terminal end of the slot. In one embodiment, the first perforation set is at the terminal end of the slot and parallel to the dressing slot axis.

(9) In one embodiment, the dressing further comprises a second perforation set through the body layer. In one embodiment, the second perforation set extends from the perimeter to within the body layer. In one embodiment, the first perforation set has a first end and a second end, and the second perforation set has a first end and a second end. In one embodiment, the first end of the first perforation set is the same as the first end of the second perforation set. In one embodiment, the first end of the first perforation set is different from the first end of the second perforation set. In one embodiment, the first perforation set is substantially perpendicular to the dressing slot axis. In one embodiment, the second perforation set is substantially perpendicular to the dressing slot axis. In one embodiment, the first perforation set extends from the first end at the perimeter through the body layer to the perimeter. In one embodiment, the first perforation set extends from the first end at the perimeter through the body layer to a different portion of the perimeter. In one embodiment, the first perforation set is at the terminal end of the slot, parallel with the dressing slot axis, and extends through the body layer to a different portion of the perimeter.

(10) In one embodiment, a method of securing a device comprises applying the dressing to a device, wherein the device aligns with the slot. In one embodiment, the method further comprises

separating at least a portion of the first perforation set adjacent to the device. In one embodiment, the method further comprises applying a tape strip across the slot, underlying the device.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a first embodiment of a dressing comprising a first perforation set and second perforation set;
- (2) FIG. 2 is a side sectional view of FIG. 1 through line 2-2;
- (3) FIG. 3 is the dressing of FIG. 1 with an underlying device at the slot and tape strip across the slot underlying the device;
- (4) FIG. 4 is another embodiment of a dressing comprising a first perforation set;
- (5) FIG. 5 is another embodiment of a dressing, specifically a tape, comprising a single perforation set;
- (6) FIG. 6 is another embodiment of a dressing, specifically a tape, comprising a first perforation set and a second perforation set;
- (7) FIG. 7 is another embodiment of a dressing comprising a first perforation set.
- (8) While the above-identified drawings and figures set forth embodiments of the invention, other embodiments are also contemplated, as noted in the discussion. In all cases, this disclosure presents the invention by way of representation and not limitation. It should be understood that numerous other modifications and embodiments can be devised by those skilled in the art, which fall within the scope and spirit of this invention. The figures may not be drawn to scale.

DETAILED DESCRIPTION

- (9) The disclosed dressing comprises a body layer with a slot for receiving a device. From the perimeter at the slot is at least one perforation set that allows for flexible expansion of the slot to accommodate various sizes of devices, such as tubing.
- (10) FIG. 1 is a first embodiment of a dressing **100** comprising a first perforation set **150** and a second perforation set **155**, FIG. 2 is a side sectional view of the dressing **100** shown in FIG. 1, and FIG. 3 is the dressing **100** shown with an underlying device **190** and an extending tube **195**.
- (11) The dressing **100** has a first major surface **110** and second major surface **120**, which is opposite the first major surface **110**. At least a portion of the second major surface **120** comprises an adhesive **160** (see FIG. 2) for securing the dressing **100** to an underlying substrate, such as skin.
- (12) A perimeter **130** surrounds the dressing **100** and defines a body layer **140** of the dressing **100**. The body layer **140** is typically a thin, flexible material that can be applied over an irregular, contoured substrate **165**, such as skin. The body layer **140** can be a single or multilayer construction and can further comprise an additional support material **170** secured to at least a portion of the body layer **140**. Various types of suitable materials for the body layer **140** and support material **170**, if included, are described below.
- (13) The dressing **100** has a slot **145**, which is an extension of the perimeter **130** within a portion of the body layer **140**. The slot **145** is separated by a first portion **146** of the body layer **140** and a second portion **148** of the body layer **140**. The slot **145** has a width forming a gap between the first portion **146** and the second portion **148** of the body layer **140**. The slot **145** has a receiving end **142** that is open and is a separation between the first portion **146** and second portion **148** of the body layer **140**, and the slot **145** has a terminal end **144** that is the innermost extension of the slot **145** into the body layer **140**. In one embodiment, the slot **145** is linear along a slot axis **143**. However, the slot **145** can be of various sizes, shapes, width or lengths so long as it provides a separation between the first portion **146** and second portion **148** of the body layer **140**. In this embodiment, the first portion **146** and second portion **148** of the body layer **140** are symmetric about the slot axis **143**. In use, the device **190**, and tubing **195**, will extend through the slot **145**.

(14) A first perforation set **150** is at the perimeter **130** at the slot **145**. The first perforation set **150** is a set of through cuts with each cut separated from another cut by the body layer **140**. The cuts of the first perforation set **150** extend at least partially through the thickness of the body layer **140** to create an area of structural weakness of the body layer **140**. In some embodiments, the through cuts of the first perforation set **150** extend entirely through the body layer **140**. In some embodiment, the through cuts extend partially into the body layer **140**. If the body layer **140** is a multilayer construction, the through cuts could extend entirely through one layer, and not through other layers of the body layer **140**. In one embodiment, such as shown in FIG. 1, the cut of the first perforation set **150** is a slit with essentially no width, meaning no material was removed from the body layer **140** during cutting. In one embodiment, the cut of the first perforation set **150** has a width of material removed.

(15) It is understood the individual cuts of a perforation set may be the same length as one another or may be of varying lengths. For example, the length of each cut may graduate from shorter to longer or from longer to shorter. The same with the spacing between each individual cut. may be the same length as one another or may be of varying lengths. For example, the spacing between each individual cut may graduate from shorter to longer or from longer to shorter.

(16) Laterally, the first perforation set **150** extends from a first end **152** at the perimeter **130** at the slot **145** to a second end **154**. In this embodiment, the second end **154** is within the body layer **140**. The first perforation set **150** can be in a line, an array, or cluster. If the first perforation set **150** is in a line, it can be a straight, angled, or curved line.

(17) The first perforation set **150** can be positioned at or near the terminal end **144** of the slot **145** as shown in FIGS. 1 and 3. The first perforation set **150** can extend at least partially from the perimeter **130** into the body **140** at any angle relative to the dressing slot axis **143**. In the embodiment depicted in FIG. 1, the first perforation set **150** extends substantially perpendicular to dressing slot axis **143** into body **140**.

(18) The dressing **100** can have more than one perforation set. Any additional perforation sets can have features such as described for the first perforation set. As shown in the embodiment in FIG. 1-3, there is a first perforation set **150** and a second perforation set **155** at the slot **145**. The second perforation set **155** is a set of through cuts with each cut separated from another cut by the body layer **140**. Similar to the first perforation set, the second perforation set **155** extends at least partially through the thickness of the body layer **140**.

(19) Laterally, the second perforation set **155** extends from a first end **156** at the perimeter **130** at the slot **145** to a second end **158**. In this embodiment, the second end **158** is within the body layer **140**. The second perforation set **155** can be in a line, and array, or cluster. If the second perforation set **155** is in a line, it can be a straight, angled, or curved line. In the embodiment depicted in FIG. 1, the second perforation set **155** extend substantially perpendicular to dressing slot axis **143** into body **140**, opposite from the first perforation set **150**. In this embodiment, the second perforation set **155** is substantially the same size and shape as the first perforation set **150**. It is understood that first and second perforation sets can be different sizes, lengths, shapes from one another. The second perforation set **155** provides an area of weakness in the structural strength of the body layer **140** of the dressing **100** and will cause a break at the second perforation set **155** to expand the side of the slot **145** for a larger tube **195**.

(20) In one embodiment, the first perforation set **150**, and second perforation set **155**, if included, have cut lengths between 0.025 inches and 1.0 inches wide. In one embodiment, the body layer **140** between cuts is between 0.01 inches and 0.05 inches wide.

(21) In use, the dressing **100** is applied over a device **190**, like shown in FIG. 3, with a portion of the device **190**, such as tubing **195** extending through the slot **145**. The tubing **195** size can vary. Therefore, the first perforation set **150** provides an area of weakness in the structural strength of the body layer **140** of the dressing **100** and will cause a break at the first perforation set **150** to expand the side of the slot **145** for a larger tube **195**.

(22) Including both a first perforation set **150** and a second perforation set **155**, increases the ability to expand the width of the slot **145** to accommodate the device **190** or tubing **195**. In one embodiment, the dressing **100** is for placement over an intravenous catheter device **190** with tubing **195**. The intravenous device **190** is placed under the body layer **140** of the dressing **100**, and the tubing **195** extends through the slot **145**. For a large size tubing **195**, at least a portion of the first perforation set **150** and/or second perforation set **155** will expand the width of slit **145** to size of the tubing **195**.

(23) Optionally, the body layer **140** of the dressing **100** can include a support material **170**. The support material **170** secures with at least a portion of the body layer **140**. In some embodiments the body layer **140** is very flexible and elastic. The support material **170** can be provided in regions to limit the flexibility and elasticity of the body layer. The support material **170** can be provided to have a flexibility or elasticity in one direction that is different than the flexibility or elasticity in a second direction. This can further help control the overall extensibility and flexibility of the overall dressing.

(24) When included, the support material **170** is secured to the body layer **140**, through adhesive, thermal bonding, lamination, or other commonly used securement techniques. The support material **170** provides structural strength to the dressing **100**, and therefore, independently, is less elastic than the body layer **140**. In one embodiment, the support material **170**, independently, has essentially no elasticity, such that it may be unable to stretch and/or it may be unable to recover from a stretch. In one embodiment, the support material **170**, may be more elastic in one direction (machine direction) than in the cross direction (transverse direction), but overall it has less elasticity than the body layer **140**.

(25) Elasticity can be measured by any number of commonly used means for evaluating stretch and recovery of a material. In one embodiment, the body layer **140** (independent of the support material) has an elongation at break of at least 200%. In one embodiment, the body layer **140** (independent of the support material) has an elongation at break of at least 1000%. In one embodiment, the body layer **140** (independent of the support material) has an elongation at break of less than 500%. In one embodiment, the support material **170** (independent of the backing layer) has an elongation at break of at least 20%. In one embodiment, the support material **170** (independent of the backing layer) has an elongation at break of at least 100%. In one embodiment, the support material **170** has an elongation at break less than 200%.

(26) In the embodiment shown in FIGS. 1-3, a support material **170** extends adjacent the entire perimeter **130** of the dressing **100**, but is not continuous across the entire body **140**. The support material **170** is not located in the central window **142** of the dressing **100** to allow for a transparent window to view the underlying substrate **165** or device **190**. In this embodiment, the support material **170** comprises polymeric film(s), nonwoven fibrous webs, woven fibrous webs, knits, films and other backing materials, which maybe translucent or transparent. The arrangement of the body layer **140** and support material **170** is similar to a 3M Tegaderm IV Advanced Securement Dressing, which includes a nonwoven layer adjacent to a thin polyurethane film. When the support material **170** is included, the perforation sets might be included in only those portion including the support material. Typically, the stiffness or rigidity of the support material will limit the propagation of the perforation opening.

(27) The body layer **140** can be thin, flexible, elastic and a sufficiently impermeable barrier to the passage of liquids and at least some gases, which protects the covered site from external contaminants. Being elastic allows for the body layer **140** to expand, contract, stretch and recover as the underlying substrate, such as skin, moves. The support material **170** can be included in specific areas, like the perimeter **130**, to limit flexibility and stretching of the body layer **140**. Excessive stretching of the body layer **140** can cause peeling or a loose fit to the underlying device. When a support material **170** is included in the dressing **100**, expansion at the slot **145** can be significantly restricted. Therefore, the perforation sets **150**, **155** provide for isolated stretching and

expansion in the areas with limited ability to stretch due to the support material **170**. If a support material **170** is included, the perforation set **150** remains within the areas of the support material **170** to prevent further propagation of the tear.

(28) FIG. **4** is a second embodiment of a dressing **102**. Dressing **102** is substantially similar to the embodiment of the dressing **100** depicted in FIG. **1**, and so the same reference numbers are used to identify the same structural elements. However, dressing **102** has only the first perforation set **150** and does not include a second perforation set. Also, the first perforation set **150** extends parallel to dressing slot axis **143** from the apex of the terminal end **144** of slot **145**. The first perforation set **150** extends from a first end **152** at the perimeter **130** to a second end **154** within the body layer **140**. This arrangement of perforation set **150** allows for selectively tearing along the perforation set **150** to accommodate a catheter of varying sizes and shapes. If a support material **170** is included, the perforation set **150** remains within the areas of the support material **170** to prevent further propagation of the tear.

(29) FIG. **5** is a third embodiment of a dressing **103**. In this embodiment, the dressing **103** is a tape strip. The same reference numbers are used to identify the same structural elements as described in FIGS. **1-3**. This dressing **103** comprises a first perforation set **150**. In this embodiment, the first perforation set **150** extends from a first end **152** adjacent the perimeter **130** at the apex of the terminal end **144** of slot **145** at least partially through the body layer **140** to a second end **154** also on the perimeter **130** along a line substantially parallel to dressing slot axis **143**. Slot **145** is positioned in the body **140** such that the portions of the body on either side of slot **145** are symmetric about dressing slot axis **143**. Support material is not included. It is understood that the support material could be included over a portion or over the entire dressing. This arrangement of first perforation set **150** allows for selectively tearing along the dressing slot axis for improved coverage of the site around a catheter and for tearing the tape completely to aid in removal of the tape. This dressing **103** can be used in combination with a dressing such as shown in FIG. **1** or FIG. **3** to further secure the device **190**, like shown in FIG. **3**.

(30) FIG. **6** is a fourth embodiment of a dressing **104**. In this embodiment, the dressing **104** is a tape strip. The same reference numbers are used to identify the same structural elements as described in FIGS. **1-3**. This dressing **104** comprising a first perforation sets **150** and second perforation set **155**. In this embodiment, the first perforation sets **150** extend from a first end **152** adjacent the perimeter **130** proximate to the apex of the terminal end **144** of slot **145** to a second end **154** into the body layer **140** along a line substantially perpendicular to dressing slot axis **143**. The second perforation sets **155** extend from a first end **156** adjacent the perimeter **130** proximate to the apex of the terminal end **144** of slot **145** to a second end **158** into the body layer **140** along a line substantially perpendicular to dressing slot axis **143**. In this embodiment, the first perforation sets **150** and second perforation set **155** are additionally symmetrical about slot **145** and the dressing slot axis **143**. Support material is not included. It is understood that the support material could be included over a portion or over the entire dressing. This arrangement of first perforation sets **150** and second perforation set **155** allows for selectively tearing along the dressing slot axis for improved coverage of the site around a catheter and accommodation of catheters of varying sizes and shapes. This dressing **104** can be used in combination with a dressing such as shown in FIG. **1** or FIG. **3** to further secure the device **190**, like shown in FIG. **3**.

(31) FIG. **7** is an embodiment of a dressing **105**. Dressing **105** is substantially similar to the embodiment of the dressing **100** depicted in FIG. **1** and dressing **102** depicted in FIG. **4**, and so the same reference numbers are used to identify the same structural elements. However, dressing **105** has a first perforation set **150** that extends perpendicular to dressing slot axis **143** from a side of the slot **145**. The first perforation set **150** extends from a first end **152** at the perimeter **130** to a second end **154** within the body layer **140**. This first perforation set **150** has a portion that extends in a first direction and a second portion that extends in a second direction. This arrangement of first perforation set **150** form an arrow “←” shape. This arrangement of perforation set **150** allows for

selectively tearing along the perforation set **150** to accommodate a catheter of varying sizes and shapes.

(32) The dressing disclosed may be made by conventional techniques (e.g., extrusion, solvent casting, calendaring, laminating, adhesive coating, and the like) which are familiar to those skilled in the art. U.S. Pat. No. 6,685,682, the disclosure of which is herein incorporated by reference, discloses constructions and methods for making dressings with backing layers and support material. It will be understood that the perforation sets can be applied using known cutting or perforation techniques to the body layer prior to final construction of the dressing or after final construction of the dressing.

(33) Body Layer

(34) The dressings are useful to provide an impermeable barrier to the passage of liquids and at least some gases. The dressings are well suited for medical application and for securing to skin. Representative barriers for use as the body layer may include non-woven and woven fibrous webs, knits, films, foams, polymeric films. In some embodiments, a transparent substrate is desirable to allow for viewing of the underlying skin or device.

(35) In one embodiment, the body layer has high moisture vapor permeability, but generally impermeable to liquid water so that microbes and other contaminants are sealed out from the area under the substrate. One example of a suitable material is a high moisture vapor permeable film such as described in U.S. Pat. Nos. 3,645,835 and 4,595,001, the disclosures of which are herein incorporated by reference. In high moisture vapor permeable film/adhesive composites, the composite should transmit moisture vapor at a rate equal to or greater than human skin such as, for example, at a rate of at least 300 g/m²/24 hrs at 37° C./100-10% RH, or at least 700 g/m²/24 hrs at 37° C./100-10% RH, or at least 2000 g/m²/24 hrs at 37° C./100-10% RH using the inverted cup method as described in U.S. Pat. No. 4,595,001. Pattern coated adhesives may be used to increase the moisture vapor transmission. In one embodiment, the body layer is an elastomeric polyurethane, polyester, or polyether block amide films. These films combine the desirable properties of resiliency, elasticity, high moisture vapor permeability, and transparency. A description of this characteristic of body layers can be found in issued U.S. Pat. Nos. 5,088,483 and 5,160,315, the disclosures of which are hereby incorporated by reference.

(36) Commercially available examples of potentially suitable body layers may include the thin polymeric film backings sold under the trade names TEGADERM (3M Company), OPSITE (Smith & Nephew), etc.

(37) In some embodiment, it may be desirable that the body layer is relatively thin to improve conformability. For example, the backing layer may be formed of polymeric films with a thickness of 200 micrometers or less, or 100 micrometers or less, potentially 50 micrometers or less, or even 25 micrometers or less.

(38) Support Material

(39) The support material provides strength to the thin, flexible body layer. The support material therefore has more stiffness and less elasticity than the body layer. The support material may be a coating, such as an adhesive, or may be a self-supporting substrate such as another film, woven, knitted, or nonwoven fabric. For example, U.S. Pat. No. 5,088,483 discloses a permanent adhesive as a reinforcement that could be used as the support material.

(40) In one embodiment, a woven, knitted, or nonwoven fabric used as the support material. One example of nonwoven for the support material is a high strength nonwoven fabric available from E. I. DuPont de Nemours & Company of Wilmington, Delaware under the trademark Sontara. Other suitable nonwoven webs include a hydroentangled polyester fabric available from Veratec, a division of International Paper of Walpole, Mass. Another suitable nonwoven web is the nonwoven elastomeric web described in U.S. Pat. No. 5,230,701, herein incorporated by reference.

(41) Adhesive

(42) Any number of adhesives can be used with the dressing. Suitable adhesives are pressure

sensitive and in certain embodiments have a relatively high moisture vapor transmission rate to allow for moisture evaporation. Suitable pressure sensitive adhesives include those based on acrylates, urethane, hydrogels, hydrocolloids, block copolymers, silicones, rubber-based adhesives (including natural rubber, polyisoprene, polyisobutylene, butyl rubber etc.) as well as combinations of these adhesives. The adhesive component may contain tackifiers, plasticizers, rheology modifiers as well as active components including for example an antimicrobial agent.

(43) The pressure sensitive adhesives that may be used in the dressings may include adhesives that are typically applied to the skin such as the acrylate copolymers described in U.S. Pat. No. RE 24,906, particularly a 97:3 isooctyl acrylate:acrylamide copolymer. Another example may include a 70:15:15 isooctyl acrylate:ethyleneoxide acrylate:acrylic acid terpolymer, as described in U.S. Pat. No. 4,737,410. Other potentially useful adhesives are described in U.S. Pat. Nos. 3,389,827; 4,112,213; 4,310,509; and 4,323,557. Inclusion of medicaments or antimicrobial agents in the adhesive is also contemplated, as described in U.S. Pat. Nos. 4,310,509 and 4,323,557.

(44) Silicone adhesive can also be used. Generally, silicone adhesives can provide suitable adhesion to skin while gently removing from skin. Suitable silicone adhesives are disclosed in PCT Publications WO2010/056541 and WO2010/056543, the disclosure of which are herein incorporated by reference.

(45) The pressure sensitive adhesives may, in some embodiments, transmit moisture vapor at a rate greater to or equal to that of human skin. While such a characteristic can be achieved through the selection of an appropriate adhesive, it is also contemplated that other methods of achieving a high relative rate of moisture vapor transmission may be used, such as pattern coating the adhesive on the backing, as described in U.S. Pat. No. 4,595,001. Other potentially suitable pressure sensitive adhesives may include blown-micro-fiber (BMF) adhesives such as, for example, those described in U.S. Pat. No. 6,994,904.

(46) Different portions of the dressing may include different adhesives, such as disclosed in U.S. Patent Application Publication 2015/014949 titled "Medical Dressing with Multiple Adhesives." For example, a portion may include an acrylate adhesive while another portion may include a silicone adhesive. In one embodiment, to prevent edge separation, adjacent the perimeter is acrylate adhesive, while near the central portion there is silicone adhesive. In one embodiment, to strongly secure with a device or tubing near the central portion there is acrylate adhesive, while near the perimeter in contact with skin is silicone adhesive.

(47) Optional Components

(48) In some applications, an absorbent material may also be used in conjunction with the dressings described herein. An absorbent material can be any of a variety of materials including, but not limited to, foam, sponge, super absorbent fibers/particles, hydrocolloid compositions, knitted, woven or nonwoven fabrics. The absorbent material can absorb body fluid underlying the dressing.

(49) An optional release liner may be included that covers all or a portion of the adhesives to prevent contamination of the adhesives prior to use. In one embodiment, the package that contains the dressing may serve as a release liner. Suitable release liners can be made of Kraft papers, polyethylene, polypropylene, polyester or composites of any of these materials. In one embodiment, the liners are coated with release agents such as silicones.

(50) An optional carrier may be included that covers all or a portion of the first major surface of the body layer, providing structural support if the dressing is thin and highly flexible. The carrier may be removable from the first major surface once the dressing is placed on skin. The carrier can be constructed of a variety of materials such as fabric that are woven or knitted, nonwoven material, papers, or film. In one embodiment, the carrier is along the perimeter of the first major surface of the dressing and is removable from the first major surface, similar to the carrier used in the 3M Tegaderm™ Transparent Film Dressing, available from 3M Company, St. Paul, MN.

(51) An optional antimicrobial component may be included that is either separate from the adhesive dressing or may be integral with the dressing. The antimicrobial component is placed near or

adjacent to the insertion site of the device **190** to inhibit microbial growth in and around the insertion site. The antimicrobial component can be absorbent foam or gel, such as used in a 3M Tegaderm™ CHG I.V. Securement Dressing, available from 3M Company.

(52) Although specific embodiments have been shown and described herein, it is understood that these embodiments are merely illustrative of the many possible specific arrangements that can be devised in application of the principles of the invention. Numerous and varied other arrangements can be devised in accordance with these principles by those of skill in the art without departing from the spirit and scope of the invention. The scope of the present invention should not be limited to the structures described in this application, but only by the structures described by the language of the claims and the equivalents of those structures.

Claims

1. A dressing comprising: a body layer including a first major surface comprising an adhesive, and a second major surface opposite the first major surface, the body layer defining a first perimeter portion; a slot defined in the body layer by a second perimeter portion of the body layer and extending from the first perimeter portion, wherein the slot extends along a slot axis from the first perimeter portion, and wherein the slot is defined by a receiving end and a terminal end in the body layer; and a first perforation set extending from the second perimeter portion at a location between the receiving end and the terminal end, wherein the first perforation set comprises: a first perforation portion extending in a first direction and terminating at an end point before reaching the first perimeter portion; a second perforation portion extending from the end point of the first perforation portion in a second direction different from the first direction.
2. The dressing of claim 1, wherein the body layer further comprises a support material.
3. The dressing of claim 2, wherein the support material covers a portion of the body layer.
4. The dressing of claim 2, wherein the support material extends to the first perimeter portion and the second perimeter portion.
5. The dressing of claim 2, the support material is less elastic than the body layer.
6. The dressing of claim 2, wherein the first perforation set is a plurality of cuts that each extend through the support material and at least partially extend through the body layer.
7. The dressing of claim 1, further comprising a second perforation set extending at least partially through the body layer.
8. The dressing of claim 7, wherein the second perforation set extends from the second perimeter portion.
9. The dressing of claim 7, wherein the first perforation set has a first end and a second end, and the second perforation set has a first end and a second end.
10. The dressing of claim 1, wherein the second perimeter portion defines: a first width of the slot at the receiving end thereof; and a second width of the slot between the terminal and receiving ends thereof, wherein the second width is greater than the first width.
11. The dressing of claim 10, wherein the second perimeter portion defines a bulbous portion of the slot having the second width, and wherein the first perforation set extends from the bulbous portion.
12. The dressing of claim 1, wherein the body layer comprises one or more layers.
13. The dressing of claim 1, wherein the slot separates first and second portions of the body layer that are symmetrical about the slot axis.
14. The dressing of claim 1, wherein the first perforation set is a plurality of cuts that each extend entirely through the thickness of the body layer.
15. The dressing of claim 1, wherein the body layer further comprises a window.
16. The dressing of claim 1, wherein the first perforation portion extends in the first direction perpendicular to the slot axis.

17. A system for securing a device comprising: the medical dressing of claim 1; and the device overlying a substrate; wherein the dressing overlies the device and secures to the substrate and wherein the device aligns with the slot.

18. A method of securing a device to a substrate comprising: applying the dressing of claim 1 to the device and the substrate; wherein the device aligns with the slot.
